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Vault Wealth: A White-Label Platform for Professional Cryptocurrency Portfolio Management

SÃO PAULO
2025

(COVER PAGE (mandatory item – NBR14724, item 4.2.1.1))

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Final Course Project submitted to the Institute of Technology and Leadership (INTELI), to obtain a bachelor's degree in Software Engineering.

Advisor: Prof. Jose Romualdo

SÃO PAULO
2025

(Catalog Card (mandatory item – NBR14724, item 4.2.1.1.2))

Cataloging in Publication
Library and Documentation Service
Institute of Technology and Leadership (INTELLI)
Data entered by the author.
(Cataloging record with international cataloging data, according to NBR 14724. The record will be completed later, after approval and before the final version is deposited. The completion of the cataloging record is the responsibility of the institution's library.)

GATTAI, Pedro. Vault Wealth: A White-Label Platform for Professional Cryptocurrency Portfolio Management. 2025. [62 pages]. Final Course Project (Bachelor) – Software Engineering, Institute of Technology and Leadership, São Paulo, 2025.

This work presents the development and validation of Vault Wealth, a white-label wealth management platform for cryptocurrency assets designed for investment offices and specialized consultancies. The cryptocurrency market is experiencing accelerated growth, with over 560 million global users and a market capitalization exceeding US$ 2.3 trillion, yet lacks integrated professional solutions for managers operating multiple portfolios distributed across centralized exchanges and on-chain wallets. The research identified that existing alternatives, primarily based on spreadsheets connected to external APIs, present critical limitations in standardization, scalability, and automation. The general objective was to create and validate a complete computational solution, developing a functional MVP and a business plan for market introduction. The methodology adopted agile development with short sprints, continuous validation with internal users at Vault Capital, and modular architecture prepared for microservices migration. The technological solution was implemented using React and TypeScript on the frontend, NestJS with Prisma ORM on the backend, PostgreSQL for persistence, Redis for caching, and RabbitMQ for asynchronous communication. The system includes multi-exchange data consolidation functionalities, automatic calculation of buy and sell operations, portfolio rebalancing algorithms, real-time analytical dashboards, and integration with HashiCorp Vault for secure credential management. Real-environment validation processed over 150 transactions with 99.8% uptime, demonstrating technical robustness and operational viability. The business model encompasses two monetization fronts: white-label licensing for investment offices via scalable monthly subscription, and direct B2C offering for individual investors. Market analysis indicates significant opportunity in Brazil, where 43% of the population has already invested in crypto assets, exceeding the global average of 42%. Results demonstrate that the platform meets identified needs, offering intelligent automation, multi-exchange integration, and a complete experience for all stakeholders in the cryptocurrency asset management ecosystem.

Keywords: crypto assets; portfolio management; white-label platform; fintech; software as a service.

Acknowledgments (optional item) – NBR14724, item 4.2.1.5)

Acknowledgments

First and foremost, I would like to express my deepest gratitude to my family for their unwavering support throughout this journey. Moving from my hometown to São Paulo to pursue this dream was only possible because of their belief in me, even during the most challenging moments of my academic trajectory.

To my co-founders and partners at Vault Capital, Igor and Allan, thank you for sharing this entrepreneurial vision and for the countless hours of collaboration that transformed an idea into a real, operating business. Building Vault together has been one of the most rewarding experiences of my life.

I extend my sincere appreciation to my advisor, Prof. José Romualdo, for his guidance, patience, and valuable insights throughout the development of this project. His mentorship helped shape both the technical and business aspects of this work.

To the entire Inteli community – professors, staff, and fellow students – thank you for creating an environment that fosters innovation, leadership, and hands-on learning. The project-based methodology fundamentally changed how I approach problem-solving and collaboration.

I am also grateful to BTG Pactual for the internship opportunity that provided me with enterprise-level experience in DevOps and infrastructure, knowledge that proved invaluable in building Vault Wealth's technical foundation.

To all the clients who trusted Vault Capital with their investments, your feedback and confidence were essential for validating and refining this platform.

Finally, to everyone who believed in me when the path was uncertain – this achievement is also yours.

Epigraph (optional item – NBR14724, items 4.2.1.6 and 5.2.4)

"If you're going through hell, keep going."

Winston Churchill

Resumo (mandatory item – NBR 14724, item 4.2.1.7)

Resumo

GATTAI, Pedro. Vault Wealth: A White-Label Platform for Professional Cryptocurrency Portfolio Management. 2025. [número de folhas]. TCC (Graduação) – Curso de Engenharia de Software, Instituto de Tecnologia e Liderança, São Paulo, 2025.

Este trabalho apresenta o desenvolvimento e validação da Vault Wealth, uma plataforma white-label de gestão de patrimônio em criptoativos voltada para escritórios de investimento e consultorias especializadas. O mercado de criptomoedas atravessa uma fase de crescimento acelerado, com mais de 560 milhões de usuários globais e capitalização superior a US$ 2,3 trilhões, porém carece de soluções profissionais integradas para gestores que operam múltiplos portfólios distribuídos entre corretoras centralizadas e carteiras on-chain. A pesquisa identificou que as alternativas existentes, baseadas principalmente em planilhas conectadas a APIs externas, apresentam limitações críticas de padronização, escalabilidade e automação. O objetivo geral foi criar e validar uma solução computacional completa, desenvolvendo um MVP funcional e um plano de negócios para sua introdução no mercado. A metodologia adotou desenvolvimento ágil com sprints curtas, validação contínua com usuários internos da Vault Capital, e arquitetura modular preparada para migração a microsserviços. A solução tecnológica foi implementada utilizando React e TypeScript no frontend, NestJS com Prisma ORM no backend, PostgreSQL para persistência, Redis para cache, e RabbitMQ para comunicação assíncrona. O sistema inclui funcionalidades de consolidação de dados multi-corretora, cálculo automático de operações de compra e venda, algoritmos de rebalanceamento de portfólios, dashboards analíticos em tempo real, e integração com HashiCorp Vault para gestão segura de credenciais. A validação em ambiente real processou mais de 150 transações com 99,8% de uptime, demonstrando robustez técnica e viabilidade operacional. O modelo de negócio contempla duas frentes de monetização: licenciamento white-label para escritórios de investimento via assinatura mensal escalável, e oferta direta B2C para investidores pessoa física. A análise de mercado indica oportunidade significativa no Brasil, onde 43% da população já investiu em criptoativos, superando a média global de 42%. Os resultados demonstram que a plataforma atende às necessidades identificadas, oferecendo automação inteligente, integração multi-corretora, e experiência completa para todos os stakeholders do ecossistema de gestão de criptoativos.

Palavras-chave: criptoativos; gestão de portfólios; plataforma white-label; fintech; software como serviço.

ABSTRACT (mandatory item – NBR 14724, item 4.2.1.8)

Abstract

GATTAI, Pedro. Vault Wealth: A White-Label Platform for Professional Cryptocurrency Portfolio Management. 2025. [number of pages]. Final Course Project (Bachelor) – Software Engineering, Institute of Technology and Leadership, São Paulo, 2025.

This work presents the development and validation of Vault Wealth, a white-label wealth management platform for cryptocurrency assets designed for investment offices and specialized consultancies. The cryptocurrency market is experiencing accelerated growth, with over 560 million global users and a market capitalization exceeding US$ 2.3 trillion, yet lacks integrated professional solutions for managers operating multiple portfolios distributed across centralized exchanges and on-chain wallets. The research identified that existing alternatives, primarily based on spreadsheets connected to external APIs, present critical limitations in standardization, scalability, and automation. The general objective was to create and validate a complete computational solution, developing a functional MVP and a business plan for market introduction. The methodology adopted agile development with short sprints, continuous validation with internal users at Vault Capital, and modular architecture prepared for microservices migration. The technological solution was implemented using React and TypeScript on the frontend, NestJS with Prisma ORM on the backend, PostgreSQL for persistence, Redis for caching, and RabbitMQ for asynchronous communication. The system includes multi-exchange data consolidation functionalities, automatic calculation of buy and sell operations, portfolio rebalancing algorithms, real-time analytical dashboards, and integration with HashiCorp Vault for secure credential management. Real-environment validation processed over 150 transactions with 99.8% uptime, demonstrating technical robustness and operational viability. The business model encompasses two monetization fronts: white-label licensing for investment offices via scalable monthly subscription, and direct B2C offering for individual investors. Market analysis indicates significant opportunity in Brazil, where 43% of the population has already invested in crypto assets, exceeding the global average of 42%. Results demonstrate that the platform meets identified needs, offering intelligent automation, multi-exchange integration, and a complete experience for all stakeholders in the cryptocurrency asset management ecosystem.

Keywords: crypto assets; portfolio management; white-label platform; fintech; software as a service.

List of Abbreviations and Acronyms (optional item – NBR14724, item 4.2.1.11)

List of Abbreviations and Acronyms

API	Application Programming Interface
AML	Anti-Money Laundering
AUM	Assets Under Management
B2B	Business to Business
B2C	Business to Consumer
BMC	Business Model Canvas
CAC	Customer Acquisition Cost
CI/CD	Continuous Integration / Continuous Deployment
CSS	Cascading Style Sheets
DeFi	Decentralized Finance
DTO	Data Transfer Object
KYC	Know Your Customer
LATAM	Latin America
LGPD	Lei Geral de Proteção de Dados
LTV	Lifetime Value
MFA	Multi-Factor Authentication
MVP	Minimum Viable Product
NFT	Non-Fungible Token
NPS	Net Promoter Score
OAuth	Open Authorization
ORM	Object-Relational Mapping
PIPE	Programa de Pesquisa Inovativa em Pequenas Empresas
RBAC	Role-Based Access Control
RLS	Row-Level Security

ROI	Return on Investment
SaaS	Software as a Service
SAM	Serviceable Available Market
SOC 2	Service Organization Control 2
SOM	Serviceable Obtainable Market
SWOT	Strengths, Weaknesses, Opportunities, Threats
TAM	Total Addressable Market
UI	User Interface
UX	User Experience

Summary (mandatory item – NBR14724, item 4.2.1.13; NBR 6027)

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1 Introduction (For titles and numerical indicators, see NBR 14724, item 5.2.2; for progressive numbering, see NBR 6024)

1.1 Context and Motivation:

The cryptocurrency market has demonstrated accelerated growth on a global scale, surpassing other technological innovations such as the internet and mobile phones in adoption speed. According to data from the "State of Global Cryptocurrency Ownership in 2024" report by Triple-A, approximately 6.8% of the world population, equivalent to more than 560 million people, already owns some type of cryptocurrency. This exponential growth is driven by several factors, including economic digitalization, which has impacted all sectors including finance, making crypto assets an increasingly accessible alternative, and the declining trust in traditional institutions, as governments, banks, and conventional monetary systems have lost credibility, especially during periods of high inflation. According to a survey by New World Wealth in partnership with Henley & Partners, the market currently has 172,300 millionaires with exposure to crypto assets, representing a 95% growth in just one year. Additionally, there are 325 centi-millionaires (with assets above US$ 100 million), an increase of 79%, and 28 billionaires directly exposed to the crypto market, a number that rose by 27% in the same period. The total number of Bitcoin users reached 275 million, a 31% growth, and its market value surpassed US$ 1.2 trillion, doubling (+103%) in just 12 months. In Brazil, the growth potential of this market is significant. A survey conducted by ConsenSys in partnership with YouGov revealed that 96% of Brazilians have heard of cryptocurrencies, and 43% of them stated they have already invested or own crypto assets, exceeding the global average of 42%. For context, the traditional investment market, represented by B3, currently has 19.4 million registered investors, demonstrating a solid and growing financial environment. This suggests significant potential for solutions specialized in cryptocurrency investment management as more investors diversify their portfolios to include digital assets. Despite this expansion, the market lacks integrated professional solutions for managers operating multiple portfolios distributed across centralized exchanges and on-chain wallets. Investment offices and consultancies that work with crypto assets face significant operational challenges, relying primarily on fragmented tools such as spreadsheets connected to external APIs, which present critical limitations in standardization, scalability, and automation.

1.2 Problem Definition and Value Proposition:

Investment offices and consultancies specializing in cryptocurrency assets face a critical operational problem: the lack of unified, professional-grade tools for managing multiple client portfolios distributed across various exchanges and on-chain platforms. The current alternatives present significant limitations including lack of standardization between different spreadsheets generating inconsistencies in analyses, high incidence of human errors in data recording and processing, difficulty in consolidating information from multiple exchanges and on-chain platforms, limitations in automating processes such as portfolio rebalancing, compromised scalability as the number of clients increases, and difficulty maintaining historical records of changes and portfolio evolution analysis. This fragmentation represents a strategic opportunity for Vault Wealth, which differentiates itself by offering an integrated and specialized system that precisely solves the main challenges of investment offices: intelligent automation, multi-exchange integration, real-time analysis, and scalability without proportional team increase, establishing significant barriers for potential competitors in a rapidly expanding market segment. The Value Proposition of Vault Wealth is to position itself as the leading white-label platform for professional cryptocurrency asset management by investment offices. In a market still marked by improvised solutions, Vault offers a reliable, scalable, and specialized infrastructure, allowing small and medium-sized businesses to operate with the same level of efficiency as large asset managers. The platform's positioning pillars include specialization in the crypto universe with mastery of market complexities through solutions designed to handle volatility, data dispersion, and on-chain integration; complete and customizable experience with specific interfaces for each function within the office adaptable to each white-label client's structure; efficiency and automation as central proposition through liberation of operational time via reliable and robust automations reducing costs and increasing scale capacity; technological scalability with multi-tenant architecture with visual identity customization ready to support multiple offices with isolation and governance; and integration with the crypto ecosystem through native integrations with exchanges, blockchains, and custody systems creating a unique and strategic view of the entire operation.

1.3 Objectives of the Work:

General Objective:

- Create and validate a complete computational solution for professional cryptocurrency portfolio management, developing a functional MVP and a comprehensive business plan for its introduction to the market as a white-label SaaS platform.

Specific Objectives:

- Develop a functional MVP with core functionalities including multi-exchange data consolidation, automated buy/sell operation calculations, portfolio rebalancing algorithms, and real-time analytical dashboards

- Implement a secure and scalable multi-tenant architecture enabling white-label deployment for multiple investment offices with complete data isolation - Integrate with major cryptocurrency exchanges via API for automated portfolio synchronization and trade execution

- Validate the platform with internal users at Vault Capital, processing real transactions and gathering feedback for iterative improvements

- Define and validate the revenue model based on monthly subscriptions with dynamic pricing considering factors such as number of clients, AUM, data volume, and active users

- Develop a comprehensive go-to-market strategy encompassing both B2B (white-label for offices) and B2C (direct to individual investors) channels

- Prepare technical and business documentation for potential investment opportunities, including PIPE/FAPESP submission

1.4 Justification and Contributions:

Market Relevance: The cryptocurrency market represents one of the fastest-growing sectors in global finance, with a total market capitalization exceeding US$ 2.3 trillion and over 560 million users worldwide. In Brazil specifically, the high adoption rate (43% of the population) combined with a mature traditional investment ecosystem (19.4 million B3 investors) creates a unique opportunity for professional crypto asset management solutions. The recent regulatory discussions around stablecoins in the United States and Brazil's advancement with the Drex digital currency project signal increasing institutional acceptance and the need for compliant, professional-grade management tools. Technological Relevance: The solution addresses a clear technological gap in the market. While traditional wealth management platforms have evolved significantly, the cryptocurrency sector still relies heavily on fragmented, manual processes. Vault Wealth introduces enterprise-grade features to this space, including real-time WebSocket infrastructure for sub-second latency updates, secure credential management through HashiCorp Vault integration, automated trading systems with intelligent routing, and scalable multi-tenant architecture with Row-Level Security for complete data isolation. Economic Relevance: The platform enables significant operational efficiency gains for investment offices, potentially reducing up to 80% of time spent on manual operational tasks through automation of critical processes such as rebalancing and alert generation. The SaaS model with scalable pricing allows for sustainable growth with low marginal cost per new client, while the white-label approach opens multiple revenue streams through B2B licensing and potential marketplace services.

1.5 Work Structure:

This work is organized into three main chapters following the introduction. Chapter 2, Solution Development, presents the complete development process divided into six sections: Section 2.1 details the market assumptions and hypotheses that guided the project including problem, solution, and value hypotheses; Section 2.2 covers market sizing and analysis using TAM, SAM, and SOM frameworks along with customer segmentation; Section 2.3 presents the competitive analysis identifying direct and indirect competitors and establishing Vault Wealth's competitive advantages through SWOT analysis; Section 2.4 describes the technological solution including requirements, architecture, development methodology, and testing strategies; Section 2.5 presents the complete business plan encompassing the Business Model Canvas, monetization strategy, go-to-market approach, and financial projections; and Section 2.6 demonstrates the validation methodology and results including KPIs and risk mitigation plans. Chapter 3 presents the conclusion, discussing whether objectives were achieved, future projections for the venture, and final considerations. The document concludes with References listing all sources consulted and Appendices containing supporting materials.

2 [Solution Development]

This chapter presents the complete development process of Vault Wealth, from initial market assumptions and hypotheses through technological implementation and business planning. The solution was developed iteratively, with continuous validation from internal users at Vault Capital, ensuring that each feature addresses real market needs identified during the research and development phases.

2.1 [Definition of Market Assumptions and Hypotheses:]

This section details the strategic analysis and definition of market assumptions that guided the project development, as well as presenting the hypotheses that directed decision-making throughout the process. These hypotheses were formulated based on direct experience operating Vault Capital, a cryptocurrency investment consultancy, and validated through continuous interaction with the market.

2.1.1 Problem Hypothesis

The core assumption is that investment offices and consultancies operating in the cryptocurrency market face significant operational inefficiencies due to the lack of integrated, professional-grade portfolio management tools. Specifically: Primary Problem: Managers who operate multiple client portfolios distributed across centralized exchanges (such as Binance and Coinbase) and on-chain wallets face a fragmented operational reality. They must manually consolidate data from multiple sources, perform calculations in spreadsheets prone to human error, and lack real-time visibility into portfolio performance across all platforms.

Pain Points Identified:

- Time-consuming manual data consolidation from multiple exchanges and wallets

- High incidence of human errors in transaction recording and portfolio calculations

- Inability to provide clients with real-time, transparent portfolio visualization

- Lack of automated rebalancing tools adapted to cryptocurrency market volatility

- Difficulty scaling operations as client base grows without proportionally increasing team size

- No unified audit trail for compliance and reporting purposes

Willingness to Pay:

Based on Vault Capital's experience serving clients with portfolios ranging from R$ 50,000 to several million reais, these investment offices are willing to pay for solutions that demonstrably reduce operational overhead, minimize errors, and enable business scaling. The current alternatives (custom spreadsheets, generic financial tools) fail to adequately address the specific needs of cryptocurrency portfolio management.

2.1.2 Solution Hypothesis

The assumption is that a white-label SaaS platform specifically designed for cryptocurrency portfolio management represents the optimal solution for the identified problems. This hypothesis is based on the following premises:

Specialization Over Generalization: Generic financial management tools fail to address the unique characteristics of cryptocurrency markets, including 24/7 trading, high volatility, multiple custody options (exchanges vs. self-custody), and the need for on-chain data integration. A specialized solution can provide features that generic alternatives cannot. White-Label Approach: Investment offices value maintaining their brand identity with clients. A white-label solution allows offices to offer a professional, branded experience without the cost and complexity of building proprietary technology. This approach also creates strong customer retention through brand integration.

Unified Platform Architecture: By consolidating data from multiple exchanges and on-chain sources into a single platform, Vault Wealth eliminates the fragmentation problem at its root. The platform becomes the single source of truth for all portfolio data, calculations, and reporting.

Automation as Differentiator: Intelligent automation of routine tasks (rebalancing calculations, report generation, alert systems) addresses the scalability problem directly. Offices can grow their client base without proportionally increasing operational staff.

Key Solution Components:

- Multi-exchange API integration for automated data synchronization
- Real-time portfolio visualization dashboards for both managers and clients
- Automated rebalancing engine with customizable strategies
- Secure credential management through enterprise-grade vault integration
- Multi-tenant architecture ensuring complete data isolation between organizations
- Role-based access control for different user types (brokers, advisors, directors, clients)

2.1.3 Value Hypothesis

The assumption is that the proposed pricing model and revenue structure are acceptable to target customers and sustainable for the business. This hypothesis encompasses: Pricing Model Acceptance: Investment offices are accustomed to paying for professional tools on a subscription basis. The proposed pricing structure based on multiple factors (number of end clients managed, total AUM, number of internal users with system access, and data volume processed and stored) aligns with industry standards while ensuring that pricing scales with the value delivered. Value Perception: The platform's value proposition centers on operational efficiency gains and business scalability. Offices that currently spend significant time on manual processes can redirect that time to higher-value activities such as client acquisition and strategic advisory. The platform essentially enables offices to operate with enterprise-level efficiency regardless of their size.

Pricing Structure for White-Label Model:

- Base monthly subscription fee - Variable component based on number of managed client portfolios - Tiered pricing based on total AUM under management

- Additional charges for premium features (advanced analytics, custom integrations) B2C Direct Model (Future): For individual investors seeking professional-grade portfolio management tools, a simplified subscription model with accessible monthly pricing provides an additional revenue stream while building brand awareness in the broader market.

ROI Justification:

Target customers can expect positive ROI through:

- Reduction in time dedicated to manual operational tasks (estimated up to 80% time savings)

- Elimination of errors from manual data processing

- Ability to serve more clients without proportional team growth

- Improved client retention through transparent, professional reporting

- Competitive differentiation in the market through technology-enabled service

2.2 [Market Sizing and Analysis:]

This section presents the market size analysis using the TAM, SAM, and SOM framework, along with detailed customer segmentation and profiling. The analysis draws from multiple data sources including industry reports, regulatory filings, and primary research conducted during platform development.

2.2.1 Market Size (TAM, SAM, SOM):

The Total Addressable Market represents the global opportunity for cryptocurrency portfolio management solutions. The cryptocurrency market has demonstrated accelerated growth on a global scale, with key indicators showing massive expansion:

Global Cryptocurrency Adoption:

- Over 560 million cryptocurrency users worldwide (6.8% of global population)

- Total cryptocurrency market capitalization exceeding US$ 2.3 trillion

- 32% year-over-year growth in total users

- 89% increase in total market value compared to previous year High-Net-Worth

Individual Participation:

- 172,300 millionaires with cryptocurrency exposure (95% annual growth)

- 325 centi-millionaires (net worth above US$ 100 million) in crypto (79% growth)

- 28 billionaires directly exposed to cryptocurrency markets (27% growth)

- 85,400 individuals became millionaires exclusively through Bitcoin (111% annual increase)

Bitcoin Specific Metrics:

- 275 million total Bitcoin users (31% growth)

- Bitcoin market value surpassed US$ 1.2 trillion (103% increase in 12 months)

- 156 centi-millionaires hold significant Bitcoin positions (100% growth)

- 11 billionaires have direct Bitcoin exposure (83% growth)

Infrastructure Maturation:

The stablecoin supply recently surpassed US$ 219 billion, reflecting growing adoption by both investors and regulators. Tether has become the 7th largest holder of US Treasury securities, holding US$ 33.1 billion in these assets as backing for USDT. This institutional-level participation signals market maturation and increased demand for professional management tools. Regulatory Tailwinds: The coordinated movement in the US Congress to approve a legal framework for stablecoins by August 2025, supported by the cryptocurrency industry and the current administration, has potential to become a watershed moment for digital asset mainstream adoption. Unlike concerns that regulation threatens the dollar, stablecoin regulation actually strengthens it by establishing the dollar as the dominant currency in the digital era through dollar-backed, auditable, and fully convertible stablecoins.

2.2.2 Customer Segmentation and Profiling

The Serviceable Available Market focuses on the Brazilian and Latin American markets where Vault Wealth can realistically operate given language, regulatory, and operational considerations.

Brazilian Market Characteristics:

- 96% of Brazilians have heard of cryptocurrencies

- 43% have already invested or own crypto assets (exceeding 42% global average) - 19.4 million investors registered on B3 (traditional stock exchange)

Growing ecosystem of investment offices and independent advisors entering crypto The high Brazilian adoption rate combined with a mature traditional investment infrastructure creates an ideal environment for Vault Wealth. Investment professionals already familiar with portfolio management concepts are increasingly seeking tools to serve clients interested in cryptocurrency exposure. Demographic and Behavioral Trends: The cryptocurrency market is experiencing vigorous expansion driven by structural and behavioral factors shaping a new generation of investors. Younger generations, such as Millennials and Generation Z, are adopting crypto assets at significantly higher rates than previous generations like Baby Boomers. This movement consolidates a new class of wealth builders, more connected to technology and open to decentralized financial models.

Target Market Segments in Brazil:

- Cryptocurrency-focused investment consultancies
- Traditional investment offices expanding into digital assets
- Independent financial advisors (AAIs) serving crypto-interested clients
- Family offices with cryptocurrency allocations
- Cryptocurrency fund managers

2.2.3 Serviceable Obtainable Market (SOM):

The Serviceable Obtainable Market represents the realistic market share Vault Wealth can capture in its initial years of operation, considering competitive dynamics, sales capacity, and operational constraints.

Initial Target:

Investment offices and consultancies already operating in the cryptocurrency space or actively expanding into it. These organizations have immediate need for the solution and shorter sales cycles due to recognized pain points.

SOM Calculation Factors:

- Number of active cryptocurrency investment consultancies in Brazil
- Average number of clients per consultancy
- Estimated monthly subscription value per client tier
- Realistic market penetration rate in years 1-3

Growth Strategy:

Begin with direct sales to known networks (Vault Capital's existing relationships), expand through referral partnerships with exchanges and custody providers, and leverage content marketing to establish thought leadership in the Brazilian crypto investment management space.

2.2.4 Customer Segmentation and Profiling

Current Users/Clients (Internal Validation Phase):

The Vault Wealth platform is currently in internal validation phase, being used by Vault Capital itself. This internal usage enables continuous development and adjustments based on real feedback from the following user groups:

- Broker Team: Uses the system daily to manage client information, register assets under management, compute buy and sell orders, and control portfolio balances
- Investment Team: Uses the platform for performance analysis, benchmark comparison, rebalancing studies, and management of company investment assets
- Investment Advisors: Employ the system to monitor client data, visualize complete portfolios, and generate personalized reports
- Directors: Access strategic information such as total invested value, current total value, recent deposits, number of new clients, and performance reports
- Clients with Managed Portfolios: Investors with Vault-managed cryptocurrency portfolios who use the client area to track investment performance, visualize current portfolio composition, and access transaction history

Future Target Clients:

Primary Target - Wealth Management Firms / Investment Advisory Offices: These are the primary candidates to become strategic Vault Wealth clients, as they share exactly the same needs that led to the platform's internal development: operational process automation, efficient multi-portfolio management, data consolidation from different exchanges, and delivery of complete end-client experience. By adopting Vault Wealth as a white-label solution, these offices can leverage all functionalities already validated and used by the Vault team, including analytical dashboards, automatic rebalancing calculation, personalized reports, and on-chain integrations, with the additional advantage of maintaining their own brand and scaling operations without increasing structure.

Secondary Target - Cryptocurrency Fund Managers: Professionals responsible for administering cryptocurrency-dedicated funds who need detailed performance analyses, rebalancing tools, and data consolidation for more effective strategic decisions.

Tertiary Target - Individual Investors and Autonomous Agents: Individuals who study the cryptocurrency market independently and maintain assets across multiple exchanges but face difficulties consolidating and efficiently managing their investments. These users value tools offering unified view of dispersed assets, performance analyses, and facilitation of tax declaration processes.

Current Commercial Partners:

- Traditional Investment Market Offices: Companies already established in traditional financial markets expanding their operations to include crypto assets. These partners function as Vault Wealth distribution channels, referring clients from their base who have interest in cryptocurrencies

- Autonomous Investment Advisors (Individual Practitioners): Independent agents acting as bridge between investors and market opportunities. The platform offers these advisors a dedicated dashboard clearly showing referred client portfolio performance and exact commission values received per referral

Future Potential Partners:

- Cryptocurrency Exchanges: Could establish partnerships to offer Vault Wealth platform as premium service to their institutional clients, creating a more complete investment experience

- Digital Asset Custody Providers: Companies specialized in securely custodying crypto assets that could benefit from integrations allowing visualization and management of custodied assets through the platform

- Specialized Accounting Firms: Firms providing accounting and tax services related to crypto assets that could use platform data to facilitate client tax declaration and accounting

2.3 [Competitive Analysis and Differentials:]

This section presents the analysis of the market and business environment, including identification of competitors and the differentiators that establish Vault Wealth's competitive advantage. The competitive landscape for cryptocurrency portfolio management solutions reveals a significant gap in the market, particularly for investment offices managing multiple client portfolios simultaneously.

2.3.1 Direct Competition

Currently, the market lacks specialized solutions for cryptocurrency portfolio management, especially for investment offices. The available alternatives present significant limitations:

Spreadsheets Connected to External APIs: This represents the most common solution, yet presents critical problems including lack of standardization between different spreadsheets generating inconsistencies in analyses, high incidence of human errors in data recording and processing, difficulty consolidating information from multiple exchanges and on-chain platforms, limitations in automating processes such as portfolio rebalancing, compromised scalability as client numbers increase, and difficulty maintaining change history and portfolio evolution analysis.

Proprietary Exchange Management Solutions: Limited to assets held on that specific platform, these solutions do not offer consolidated view of investments distributed across multiple exchanges or on-chain wallets. They frequently lack advanced analysis and portfolio management tools, focusing primarily on trading functionalities. This limitation makes them completely inadequate for the professional context of investment offices needing to manage diverse client portfolios with assets across multiple platforms.

Monitoring Dashboards and Tracking Applications: Various solutions exist in the market providing price visualization, trends, and basic portfolio tracking. However, they present significant limitations as these tools are primarily designed for individual retail investors managing only a single portfolio. They serve only for monitoring purposes, as transaction operationalization still needs to be performed directly on exchanges or on-chain, creating a fragmented experience. They lack functionalities essential for professional management such as multiple client portfolio control, commission calculation and tracking, automatic rebalancing tools, and advanced analysis resources. The competitive analysis reveals a critical gap in the market for cryptocurrency portfolio management solutions, especially for offices managing multiple portfolios simultaneously.

Existing alternatives present significant limitations: spreadsheets with APIs suffer from standardization and scalability problems; generalist financial software does not address crypto market particularities; proprietary exchange solutions are restricted to their own platforms; and monitoring dashboards serve only basic tracking purposes.

2.3.2 Indirect Competition

As indirect competition, the following alternatives can be considered:

Traditional Financial Management Software: Platforms designed for traditional asset management that some offices attempt to adapt for cryptocurrency use. These solutions lack native integrations with cryptocurrency exchanges, do not understand crypto-specific concepts such as wallet addresses, on-chain transactions, or DeFi protocols, and require significant manual workarounds to function in the crypto context.

Custom Internal Development: Some larger investment offices attempt to build proprietary solutions internally. This approach requires significant upfront investment, ongoing development resources, and technical expertise that most small to medium offices lack. The build-versus-buy analysis typically favors specialized SaaS solutions for all but the largest organizations.

General Portfolio Tracking Apps (Retail-Focused): Applications like CoinGecko Portfolio, Delta, or Blockfolio that allow individuals to track cryptocurrency holdings. While useful for personal use, these lack multi-client management capabilities, professional reporting features, white-label options, and the security/compliance features required by investment offices.

Manual Processes: The status quo for many offices remains a combination of exchange interfaces, spreadsheets, and manual record-keeping. While functional at small scale, this approach becomes increasingly untenable as client numbers grow and regulatory requirements increase.

2.3.3 Competitive Advantages

Vault Wealth has developed significant competitive advantages that position it uniquely in the market:

Unified and Automated System:

- Inconsistency Elimination: Single interface that centralizes data previously dispersed across multiple systems, ensuring information consistency and providing a single source of truth for all operations

- Significant Error Reduction: Minimization of manual intervention through automated processes, drastically reducing human errors in data recording and processing that are common in spreadsheet-based solutions

- Intelligent Automation: Critical processes fully automated, including buy/sell operation calculations and portfolio rebalancing based on mathematical functions, allowing professionals to focus on higher-value strategic activities

Multi-Exchange and On-Chain Integration:

- Comprehensive Consolidation: Aggregation of asset information dispersed across different exchanges and on-chain platforms, eliminating the need to access multiple systems to obtain complete investment view

- Holistic Vision: Complete panorama of total managed assets, facilitating comparative analysis, data-based decision making, and real-time opportunity identification

- Unified Tracking: Centralized asset monitoring independent of custody location, providing total transparency over movements and performance in a single environment

Cryptocurrency Specialization:

- Adapted Algorithms: Developed specifically to handle the characteristic volatility of cryptocurrency markets, offering analyses and recommendations that consider this sector's particularities

- Domain Knowledge: Incorporation of expertise from professionals with extensive crypto market experience, resulting in functionalities that address real portfolio manager needs

- Proprietary Methodology: Unique approach for crypto portfolio analysis and management, developed internally based on Vault team's practical experience in digital asset management

Scalable Architecture:

- Microservices Infrastructure: Modern architecture designed to grow as managed portfolios increase, enabling scale without performance degradation

- Frictionless Growth: Capacity to expand managed client numbers without proportional operational team increase, resulting in scale economics and greater profitability

- Technological Adaptability: Flexibility to incorporate new functionalities, exchanges, and integrations with ease, ensuring the platform continuously evolves with the market Complete Stakeholder Experience:

- Personalized Access Levels: Efficient delegation of responsibilities among different team functions, ensuring each professional has access to exactly the tools and information necessary for their role

- Intuitive Client Portal: Friendly interface for investors to monitor their assets, promoting transparency and strengthening trust relationship between clients and consultants

2.3.4 SWOT Analysis

Strengths:

- Intelligent Automation of Critical Processes: Reduction in time dedicated to operational tasks, elimination of human errors in data recording and processing, capacity to scale operations without proportional team increase

- Comprehensive Multi-Exchange and On-Chain Integration: Native connections with major global market exchanges, real-time monitoring of performed operations, unified view of assets dispersed across different platforms

- Advanced Real-Time Analysis: Dashboards providing detailed and real-time performance metrics to track portfolios, market, and company

- Institutional Expertise in Cryptocurrency Market: Rebalancing algorithms developed based on real market knowledge, proprietary analysis methodology adapted to sector peculiarities

- Modern and Scalable Technology Architecture: Technology stack focused on maintainability, security, and performance, modular structure prepared for gradual migration to microservices, scalable cloud architecture

Weaknesses:

- Limited External Validation Base: Product validated primarily internally, with limited exposure to external clients, need to evaluate performance and usability at larger scale, potential gap between perceived needs and real market needs

- Interface Refinement Need: UX/UI in evolution process to reach competitive market standards, opportunity to improve learning curve for new users, need for optimization for different devices

- Limited Financial Resources for Aggressive Marketing: Competition with established traditional financial market players, need for significant investment for brand awareness, potentially long sales cycle for B2B products in financial sector

- Development Team in Expansion Stage: Need to grow technical team to accelerate deliveries, challenge of maintaining code quality during product scaling, requirement for specialists in specific areas

Opportunities:

- Accelerated Growth Cryptocurrency Market: Growing global adoption with more than 560 million users, institutional investor entry into crypto market, market maturation creating demand for professional solutions

- Gap in Specialized Market Solutions: Absence of integrated platforms for crypto portfolio managers, operational difficulties faced by offices trying to enter the sector, demand for tools ensuring regulatory conformity

- International Market Expansion: Potential for product adaptation to other Latin American markets, opportunity for partnerships with global financial institutions, growing demand for solutions in emerging markets

- Potential for Additional Revenue Models: Complementary services marketplace, monetization of aggregated and anonymized data (with consent), API offering for third-party developers

Threats:

- Entry of Established Financial Market Players: Large financial institutions with resources to develop proprietary solutions, well-funded fintechs directing efforts to crypto segment, competition for scarce talent in finance and blockchain

- Cybersecurity Risks: Financial sector as preferential attack target, need to maintain constantly updated security measures, client concerns with data protection and access to sensitive information

- Rapid Cryptocurrency Market Regulation Evolution: Regulatory changes may require significant product adaptations, requirement variations between different jurisdictions, compliance as constant challenge in evolving regulatory environment

2.4 [Technological Solution]

This section presents the technological and computational solutions developed for Vault Wealth, including system requirements, architecture decisions, development methodology, and testing strategies. The platform evolved from an internal tool at Vault Capital into a comprehensive white-label solution through iterative development and continuous validation.

2.4.1 Requirements and Specifications:

Functional Requirements:

Portfolio Management:

- FR01: System shall consolidate portfolio data from multiple cryptocurrency exchanges via API integration

- FR02: System shall support on-chain wallet tracking for major blockchain networks

- FR03: System shall calculate portfolio performance metrics in real-time

- FR04: System shall provide automated rebalancing recommendations based on configurable strategies

- FR05: System shall execute buy/sell orders through integrated exchange APIs

- FR06: System shall maintain complete transaction history with audit trails

User Management:

- FR07: System shall support multi-tenant architecture with complete data isolation between organizations

- FR08: System shall implement role-based access control (RBAC) with customizable permission hierarchies

- FR09: System shall provide different interface views for different user types (brokers, advisors, directors, clients)

- FR10: System shall support white-label customization including branding, colors, and logos per tenant

Reporting and Analytics:

- FR11: System shall generate customizable performance reports in multiple formats (PDF, Excel)

- FR12: System shall provide real-time analytical dashboards with key performance indicators

- FR13: System shall support benchmark comparison against market indices

- FR14: System shall track and display commission calculations for referral partners

Communication and Notifications:

- FR15: System shall provide real-time notifications for important portfolio events via WebSocket

- FR16: System shall support configurable alerts for price movements, rebalancing triggers, and system events

- FR17: System shall maintain notification history and read/unread status

Non-Functional Requirements:

Performance:

- NFR01: System shall maintain sub-second latency for real-time data updates via WebSocket

- NFR02: System shall support concurrent connections from 1000+ users without performance degradation

- NFR03: API response times shall not exceed 200ms for standard operations under normal load

Security:

- NFR04: System shall encrypt all data in transit using TLS 1.3
- NFR05: System shall encrypt sensitive data at rest using AES-256
- NFR06: System shall store exchange API credentials in HashiCorp Vault with secure retrieval
- NFR07: System shall implement Row-Level Security (RLS) in PostgreSQL for tenant data isolation
- NFR08: System shall maintain comprehensive audit logs for all data modifications

Availability and Reliability:

- NFR09: System shall maintain 99.8% uptime availability
- NFR10: System shall implement automated backup with point-in-time recovery capability
- NFR11: System shall support graceful degradation when external services are unavailable

Scalability:

- NFR12: Architecture shall support horizontal scaling of application servers
- NFR13: Database shall support read replicas for query distribution
- NFR14: System shall be designed for future migration to microservices architecture

User Specifications and Use Cases:

Broker Team Use Cases:

- Register new client portfolios and configure exchange API connections
- Record deposits and withdrawals across client accounts
- Execute buy/sell orders and track order status
- Monitor daily portfolio balances and alert conditions

Investment Advisor Use Cases:

- View consolidated portfolio data across all assigned clients
- Generate performance reports for client meetings
- Configure rebalancing strategies and review recommendations
- Add contextual notes to client portfolios

Director/C-Level Use Cases:

- Access company-wide analytics dashboard
- Monitor total AUM, revenue metrics, and growth indicators
- Review new client acquisition and churn data
- Generate compliance and audit reports

End Client Use Cases:

- View personal portfolio performance and composition
- Access transaction history and statements
- Download tax-related reports
- Receive notifications for significant portfolio events

2.4.2 Architecture and Technology:

System Architecture:

Vault Wealth employs a modular monolithic architecture designed for future migration to microservices. This hybrid approach enables rapid development in initial phase, gradual decomposition into independent services as complexity increases, technological flexibility allowing choice of most suitable stack for each domain, and independent scalability of components according to demand.

Current Architecture Components:

- Main Service: Core module responsible for the majority of software business logic
- Trade Service: Module responsible for integration with exchanges for buy and sell operations
- Data Scraping Service: Module for web scraping market data and PDF report generation
- Frontend: Module for user visualization and interaction with the system
- Database: Relational database for data storage
- Cache: Module responsible for temporary storage of frequently accessed data
- Message Broker: Module used for asynchronous communication between services

Technology Stack:

Frontend:

- React.js with TypeScript for type-safe, component-based UI development
- Tailwind CSS for utility-first styling and responsive design
- WebSocket client for real-time data updates
- React PDF Viewer for interactive document visualization

Backend:

- NestJS framework for scalable, enterprise-grade Node.js applications
- TypeScript for type safety and improved developer experience
- Prisma ORM for type-safe database access and migrations
- 22 NestJS modules comprising 17,622+ lines of production code

Database and Storage:

- PostgreSQL for primary relational data storage with Row-Level Security
- Redis for caching and session management
- HashiCorp Vault for secure credential storage

Message Queue and Real-Time:

- RabbitMQ for asynchronous message processing
- WebSocket gateways for real-time notifications (4 specialized gateways)
- Sub-second latency for critical financial event updates

Infrastructure:

- Docker for containerization and deployment consistency
- CI/CD pipelines for automated testing and deployment
- Cloud-native design for scalability and reliability

Orchestration and Deployment (Future): To ensure scalability and ease in deployment, the platform will utilize Kubernetes for container/pod orchestration. Migration to microservices will be performed incrementally, prioritizing components with greater need for scalability (such as market data processing) and independent maintenance (including exchange integrations and authentication systems).

2.4.3 Development and Implementation (MVP):

Development Methodology:

The project adopted Agile development methodology with the following characteristics:

Sprint Planning: Clear organization of deliveries and priorities using Linear for project management, providing real-time visibility of development progress and better alignment between technical and business stakeholders.

Rapid Iterations: Short feedback cycles with continuous improvements, focused MVP prioritizing core functionalities for quick validation.

Technical Documentation: Comprehensive documentation platform deployed using Docusaurus, including architectural overview, API module references, Redis quick-start guide, WebSocket protocol specifications, and detailed trading flow documentation.

Development Phases:

Phase 1 - Foundation (Months 1-3):

- Core authentication system with Auth0 integration
- Basic portfolio data model and CRUD operations
- Initial exchange API integration (Binance, Coinbase)
- Fundamental dashboard UI components

Phase 2 - Multi-Tenancy (Months 4-6):

- Complete multi-tenant structure implementation with tenant_id across all tables
- Row-Level Security (RLS) PostgreSQL configuration for data isolation

- Role-Based Access Control (RBAC) with custom hierarchies
- Frontend/Backend integration with tenant-aware routing

Phase 3 - Trading Automation (Months 7-9):

- Automated trading system with exchange integration (api-transactions module)
- Intelligent decision engine for real vs. mock trade execution
- PendingTransaction model with full lifecycle management
- Asynchronous transaction infrastructure with retry mechanisms

Phase 4 - Real-Time Infrastructure (Months 10-12):

- WebSocket gateway implementation (4 specialized gateways)
- Wallet notifications gateway for portfolio updates
- Transactions gateway for trade execution streaming
- Health monitoring and missing assets detection

Key MVP Features Implemented:

- Multi-exchange data consolidation from Binance and Enor
- Automated buy/sell operation calculations based on mathematical functions
- Portfolio rebalancing algorithms with customizable strategies
- Real-time analytical dashboards with performance metrics
- Secure credential management through HashiCorp Vault
- Client notes management system for portfolio observations
- Comprehensive audit logging for compliance requirements

2.4.4 Testing and Technical Evaluation:

Testing Strategies:

Unit Testing: Individual component testing ensuring each module functions correctly in isolation. Test coverage focuses on critical business logic including portfolio calculations, rebalancing algorithms, and transaction processing.

Integration Testing: Testing of component interactions, particularly API integrations with external exchanges, database operations with Prisma ORM, and WebSocket communication flows.

End-to-End Testing: Complete user flow testing from authentication through transaction execution, validating the entire system operates correctly as an integrated whole.

Comprehensive Testing Suite: Three test suites totaling 634+ lines of test code covering deposit/withdrawal flows (depositWithdrawal.spec.ts), focused scenarios (focused.spec.ts), and simple operations (simple.spec.ts), achieving significant coverage of critical trading paths and edge cases.

Technical Evaluation Results:

Wallet Consistency Validation Framework: The walletConsistencyValidator.service (465 lines) ensures data integrity during complex trading operations, validating asset quantities, allocation percentages, and total portfolio values before and after transactions, preventing calculation errors and maintaining audit trail compliance.

Production Testing Results:

- Platform processed 150+ real transactions across 2 different exchanges
- Achieved 99.8% uptime during 1-month production testing period
- Intelligent routing system automatically selects best execution venue based on liquidity and fees
- Mock trading feature validated for strategy testing without capital risk

Performance Metrics:

- Sub-second latency achieved for WebSocket real-time updates
- API response times consistently under 200ms for standard operations
- Successful handling of concurrent connections from multiple users
- Database query optimization reduced N+1 problems by 70% through Prisma include statements

Security Validation:

- HashiCorp Vault integration validated for secure credential storage and retrieval
- Row-Level Security confirmed for complete tenant data isolation
- Auth0 integration tested for OAuth 2.0 and MFA authentication flows
- Preliminary compliance achieved with CCSS (Cryptocurrency Security Standard)

2.5 [The Business Plan]

This section presents the comprehensive business plan for Vault Wealth, including the Business Model Canvas, monetization strategy, go-to-market approach, and financial projections. The plan is designed to support both the current operational phase and future scaling of the platform.

2.5.1 Business Model Canvas:

Value Propositions:

White-label cryptocurrency portfolio management platform for investment offices. Elimination of manual processes through intelligent automation. Multi-exchange and on-chain data consolidation in a single interface. Scalable operations without proportional team growth. Professional-grade tools accessible to small and medium-sized offices. Complete transparency for end clients through dedicated portal.

Customer Segments:

Primary: Wealth management firms and investment advisory offices operating in cryptocurrency markets. Secondary: Cryptocurrency fund managers requiring professional portfolio tools. Tertiary: Independent financial advisors expanding into digital assets. Future: Individual investors seeking professional-grade portfolio management (B2C).

Channels:

Direct sales to investment offices through consultative selling approach. Strategic partnerships with cryptocurrency exchanges. Content marketing and thought leadership (webinars, whitepapers, blog). Industry events and conferences in financial and crypto sectors. Referral program for existing clients and partners.

Customer Relationships:

Dedicated onboarding and implementation support. Ongoing technical support and customer success management. Regular product updates and feature releases based on feedback. Community building through user groups and events (Crypto Sessions by Vault). Educational content and training materials.

Revenue Streams:

Monthly SaaS subscription fees (white-label licensing). Variable pricing based on AUM, client count, and user seats. Premium feature add-ons (advanced analytics, custom integrations). Future: B2C direct subscriptions for individual investors. Future: Marketplace commissions for complementary services.

Key Resources:

Proprietary technology platform and intellectual property. Development team with expertise in fintech and blockchain. Domain knowledge in cryptocurrency investment management. Strategic relationships with exchanges and industry partners. Vault Capital as internal validation and reference customer.

Key Activities:

Continuous platform development and feature enhancement. Exchange API integration maintenance and expansion. Customer acquisition and sales process execution. Customer success and support operations. Security maintenance and compliance monitoring. Market research and competitive intelligence.

Key Partnerships:

Cryptocurrency exchanges (Binance, Coinbase, Kraken) for API access. Authentication providers (Auth0) for identity management. Cloud infrastructure providers for hosting and scalability. Security vendors (HashiCorp) for credential management. Potential: Custody providers, accounting firms, educational platforms.

Cost Structure:

Personnel costs (development team, sales, support). Cloud infrastructure and hosting expenses. Third-party service subscriptions (Auth0, monitoring tools). Marketing and customer acquisition costs. Legal and compliance expenses. Administrative and operational overhead.

2.5.2 Monetization Strategy:

- **Current Model (Internal Validation Phase):**
- The platform is currently in internal use phase by Vault Wealth, allowing functionality refinement and validation based on real operations before external commercialization.
- **Future Model (SaaS White-Label and Direct B2C):**
- Vault Wealth will be commercialized as a SaaS solution in two complementary revenue models: (i) white-label for investment offices and (ii) direct subscription for individual investors.
- **White-Label Model for Investment Offices:**
- The platform will be licensed as white-label for investment offices and consultancies specialized in crypto assets, allowing brand customization, visual identity, and access levels. The pricing model is

based on monthly subscription, scaling according to platform usage by each partner.

- **Pricing Base for White-Label:**
- Number of managed end clients. Total volume of assets under management (AUM). Number of internal users with system access. Volume of data processed and stored.
- **Direct Model for Individual Investors (B2C):**
- Beyond white-label, Vault will also offer the platform directly to individual investors who wish to professionally manage their own crypto assets. This version will maintain the main functionalities of data consolidation, portfolio visualization, and performance analysis, with pricing via simple and accessible monthly subscription aimed at the end investor.
- **Main Operational Costs for the Platform:**
- Scalable cloud infrastructure. Development team and technical support. Service to white-label partners and individual users. Legal and regulatory support (compliance, KYC/AML). Marketing and client acquisition actions.
- This hybrid monetization structure maximizes platform revenue potential, serving both institutional audience and the growing market of autonomous investors.

2.5.3 Go-to-Market Strategy:

The market entry strategy for Vault Wealth will be structured in progressive phases, combining validation with strategic partners, active client acquisition, and brand strengthening in both crypto and traditional financial ecosystems. The model contemplates both B2B (white-label) and B2C (direct subscription by investors) approaches.

Phase 1: Market Validation and Proof (Controlled Pilot)

Objective: Collect real usage feedback, validate critical functionalities, and measure operational and financial impact for first institutional clients.

Actions: Launch pilots with investment offices and partner advisors already connected to Vault. Intensive monitoring of metrics such as NPS, CAC, LTV, and churn during initial phase. Rapid product iterations based on real-time feedback. Documented use cases for use in subsequent marketing.

Phase 2: B2B Acquisition Strategy (White-Label for Offices)

Acquisition Channels: Consultative direct sales to investment offices via specialized commercial team. Segmented inbound marketing with educational content production targeting managers and advisors (e.g., whitepapers, webinars, articles on crypto asset management). Financial and crypto sector events and conferences to generate authority and qualified leads.

Strategic Partnerships: Cryptocurrency exchanges that can offer Vault as white-label platform to their institutional clients. Custodians and accounting firms specialized in digital assets. Associations and innovation hubs in financial sector and Web3.

Phase 3: B2C Acquisition Strategy (Individual Investors)

Acquisition Channels: Digital growth marketing with campaigns focused on niches such as crypto investors, DeFi enthusiasts, and investors with distributed assets across exchanges. Influencer partnerships for dissemination through partnerships with crypto and traditional market influencers. Organic marketing through social media (YouTube, Instagram, X, etc.), newsletters, blogs, forums. Community incentive for recurrent platform use and new user referrals.

Value Proposition Highlight: Professional management for individual crypto investors with consolidated visualization from multiple exchanges, rebalancing automation, and reports ready for tax declaration.

Phase 4: Growth and Retention Levers

Referral program targeting advisors and offices to increase client base. Notification system used to improve user engagement. Personalized performance and compliance reports as competitive differential for institutional client loyalty.

2.5.4 Financial Projection and Feasibility:

Revenue Model and Pricing Structure:

The revenue model is based on recurring monthly subscriptions with pricing tiers designed to scale with customer growth and value delivered.

White-Label Pricing Tiers (Illustrative):

Starter: Up to 50 managed clients, basic features, R$ 2,000-5,000/month. Professional: Up to 200 managed clients, advanced analytics, R$ 5,000-15,000/month. Enterprise: Unlimited clients, full customization, custom pricing based on AUM.

B2C Direct Pricing (Future):

Individual subscription: R$ 49-199/month depending on feature tier.

Investment Opportunity with R&D Focus (PIPE/FAPESP):

Vault Wealth is a SaaS white-label platform focused on professional cryptocurrency asset management, targeting investment offices operating multiple portfolios distributed across centralized exchanges, digital wallets, and public blockchains. The solution unifies dispersed data, automates operational tasks, and offers functionalities such as rebalancing, risk analysis, and real-time reports through a modern and customizable interface.

Already validated in real environment with internal use, the platform is in transition stage to a scalable and secure architecture, with modular components enabling multi-tenant operation and fluid integration with off-chain and on-chain data sources.

PIPE Phase 1 Objective (R$ 300,000):

Phase 1 of PIPE will be dedicated to advancing the technological maturity of Vault Wealth platform, through consolidation of its scalable architecture and development of critical functionalities for multi-portfolio environment operation with security and efficiency. The main objectives of this stage are:

Complete architecture transition to microservices, with emphasis on separation of critical domains such as authentication, portfolio management, and data processing. Develop first autonomous intelligent automation modules including configurable automated rebalancing engine and event-based notification and alert system. Implement multi-origin data consolidation system with simultaneous integration of off-chain data (via exchanges like Binance and Enor). Test solution in real environment with internal users, validating usability, performance, and operational impact in investment offices operating with multiple crypto portfolios. Generate technical and commercial viability evidence, consolidating a business plan and detailed technical roadmap for Phase 2.

Resource Allocation Schedule - Phase 1 (PIPE):

Months 1-2: Scholarship holder hiring, development team structuring and architecture (R$ 50,000). Months 2-4: Architecture transition to microservices and DevOps implementation (R$ 60,000). Months 3-5: Implementation of off-chain and on-chain data consolidation layer (R$ 40,000). Months 5-6: Development of automated rebalancing engine and alert system (R$ 70,000). Months 5-6: Real environment testing with internal users and partner offices (R$ 30,000). Month 6: Technical report, updated business plan, and Phase 2 roadmap (R$ 20,000). Technical contingency reserve (R$ 30,000). Total Phase 1 PIPE: R$ 300,000.

Phase 2 Projection and Continuity (R$ 1,500,000):

If Phase 1 is successfully completed, the plan is to evolve to PIPE Phase 2, focusing on: Architecture scalability for multiple tenants (white-labels). Analysis engine optimization with predictive algorithms. Mobile application with embedded intelligence. Security and compliance certifications. Internationalization and model replication plan for LATAM markets.

Expected Impact Indicators:

Completion of architecture migration to microservices with clear domain separation and scalability and security gains. Up to 80% reduction in internal operational time through automation of critical tasks such as rebalancing and alert generation. Creation of an automated rebalancing engine and event-based alert system with customizable metrics per manager and strategy. Implementation of functional layer capable of performing buy and sell operations within the software with exchange integrations. Production of structured technical and business roadmap for Phase 2, focusing on multi-tenant scalability and international expansion. Training of at least 2 qualified scholarship holders in technologies applied to crypto asset ecosystem. Development of at least one component with intellectual property protection potential as part of intelligent automation engine.

Alignment with PIPE Objectives:

Concrete and relevant technological innovation: Vault Wealth proposes construction of a highly specialized technical core for crypto asset management, involving algorithmic automation, modular architecture, and integration of decentralized data sources.

Applied research with scientific basis: Phase 1 involves computational challenges related to scalability, domain segmentation, and construction of autonomous systems based on rules and business logic, with potential for evolution to predictive models and machine learning in Phase 2.

Direct impact on productive sector: The solution addresses a clear pain point of offices managing multiple portfolios, offering an automated and efficient alternative to the current manual and poorly scalable model.

Market insertion viability: Vault already operates with real users in controlled environment and has commercial partners interested in testing the platform in its white-label version. The SaaS model allows scalability with low marginal cost per new client.

Continuity and growth potential: The platform has a well-defined long-term roadmap, with possibility of intellectual property registration, international expansion, and connection with ecosystems such as custodians, exchanges, and data providers.

2.6 [Validation and Results]

This section demonstrates the validation of the Vault Wealth project in the market, presenting the methodology used, results obtained, key performance indicators, and risk mitigation strategies. The validation process was conducted through real-world usage rather than merely theoretical exercises, ensuring that findings reflect actual market conditions and user needs.

2.6.1 Validation Methodology:

The validation methodology for Vault Wealth combined multiple approaches to test both the business hypotheses and the MVP acceptance in real-world conditions.

Internal Production Testing:

The primary validation method involved deploying the platform for managing Vault Capital's internal cryptocurrency portfolio operations. This approach provided continuous real-world feedback in an environment where errors have actual financial consequences, ensuring rigorous testing of all functionalities.

User Feedback Collection:

Systematic feedback collection from different user types within Vault Capital including broker teams, investment advisors, directors, and clients with managed portfolios. Each user group provided insights specific to their workflow needs and pain points.

Iterative Development Cycles:

Short sprint cycles allowed rapid incorporation of user feedback into product development. Features were prioritized based on actual usage patterns and explicit user requests rather than assumptions.

Technical Validation:

Comprehensive testing suite implementation covering unit tests, integration tests, and end-to-end scenarios. Production monitoring with detailed logging enabled identification and resolution of issues in real-time.

Market Validation Activities:

Conversations with potential external clients (investment offices and independent advisors) to validate problem-solution fit. Analysis of competitor offerings and identification of feature gaps. Documentation of use cases for future marketing and sales activities.

Metrics Tracking:

Implementation of analytics and monitoring tools to track platform usage patterns, performance metrics, and user engagement indicators throughout the validation period.

2.6.2 Market Validation Results: Market Validation Results

Production Performance Data:

The platform was deployed in production environment managing real cryptocurrency portfolios at Vault Capital. Key results from this validation period include:

Platform processed over 150 real transactions across 2 different exchanges (Binance and Enor). System achieved 99.8% uptime during the 1-month intensive testing period. Zero critical errors in financial calculations or transaction processing. Successful handling of concurrent users accessing the platform simultaneously.

User Feedback Insights:

Broker Teams reported significant time savings in daily operations, particularly in portfolio balance reconciliation and order processing. The automated calculation features eliminated manual spreadsheet work that previously consumed hours daily.

Investment Advisors highlighted the value of consolidated portfolio views and customizable reports for client meetings. The ability to generate professional reports directly from the platform improved client communication quality.

Directors appreciated the strategic dashboard providing real-time visibility into total AUM, client acquisition metrics, and company-wide performance indicators.

End Clients expressed satisfaction with the transparency provided by the client portal, enabling them to track their investments without requiring direct contact with advisors for routine information.

Technical Validation Results:

WebSocket infrastructure delivered sub-second latency for real-time updates across all four specialized gateways. Intelligent trade routing system successfully determined optimal execution paths based on available credentials. Mock trading feature validated as effective tool for strategy testing without capital risk. HashiCorp Vault integration confirmed secure for credential storage and retrieval operations.

Pivoting and Persisting Decisions:

Based on validation feedback, several product decisions were made:

Persisted: Multi-tenant architecture with Row-Level Security proved essential and was fully implemented. Real-time WebSocket notifications confirmed as critical feature for user satisfaction. Automated rebalancing calculations validated as key differentiator.

Pivoted: Initial focus on extensive exchange integrations was narrowed to prioritize depth over breadth, focusing on most-used exchanges first. User interface simplified based on feedback about complexity, prioritizing core workflows over advanced features. Mobile app development deprioritized in favor of responsive web design for initial release.

Market Interest Indicators:

Conversations with external investment offices confirmed strong interest in white-label solution. Multiple potential partners expressed willingness to participate in pilot programs. Traditional investment offices expanding into crypto identified as particularly receptive audience.

2.6.3 Key Performance Indicators (KPIs):

Key Performance Indicators (KPIs)

Technical Performance KPIs:

System Uptime: 99.8% achieved during validation period, exceeding the 99.5% target for production systems.

API Response Time: Average response time under 200ms for standard operations, meeting performance requirements.

WebSocket Latency: Sub-second latency achieved for real-time notifications across all gateway types.

Transaction Processing Accuracy: 100% accuracy in financial calculations with zero discrepancies identified during audit.

Code Quality: 17,622+ lines of production code across 22 NestJS modules with 634+ lines of test coverage.

Operational Efficiency KPIs:

Time Savings: Estimated 80% reduction in time dedicated to manual operational tasks through automation of rebalancing and alert generation.

Error Reduction: Elimination of human errors in data recording and processing previously common in spreadsheet-based workflows.

Scalability: Platform architecture validated to support growth in client base without proportional increase in operational team.

User Engagement KPIs:

Daily Active Usage: Consistent daily usage by broker teams and investment advisors throughout validation period.

Feature Adoption: High adoption rates for core features including portfolio dashboard, transaction recording, and report generation.

User Satisfaction: Positive qualitative feedback from all user segments regarding platform usability and value delivered.

Business Validation KPIs:

Transaction Volume: 150+ real transactions processed successfully during validation period.

Multi-Exchange Integration: Successful data consolidation from 2 exchanges with architecture supporting additional integrations.

Client Portfolio Management: Multiple client portfolios managed simultaneously with complete data isolation confirmed.

Future KPIs for Commercial Phase:

Customer Acquisition Cost (CAC): To be measured upon commercial launch, target below industry average for B2B SaaS.

Customer Lifetime Value (LTV): Target LTV:CAC ratio of 3:1 or higher for sustainable growth.

Monthly Recurring Revenue (MRR): Growth targets to be established based on initial commercial traction.

Churn Rate: Target below 5% monthly churn for white-label clients.

Net Promoter Score (NPS): Target NPS above 50 indicating strong customer satisfaction and referral potential.

2.6.4 Risks and Mitigation Plan:

- **Financial Risks:**
- Risk: Insufficient funding to complete product development and achieve market traction before runway depletion.
- Mitigation: Pursuit of PIPE/FAPESP funding (R$ 300,000 Phase 1) to extend runway. Lean operational structure minimizing fixed costs. Revenue generation through Vault Capital operations providing baseline income. Phased development approach allowing value delivery at each stage.
- Risk: Pricing model rejection by target market resulting in slower than projected revenue growth.
- Mitigation: Flexible pricing tiers accommodating different client sizes and needs. Continuous market feedback collection to adjust pricing strategy. Value-based pricing aligned with measurable client benefits (time savings, error reduction).
- **Technological Risks:**
- Risk: Exchange API changes or deprecations disrupting platform functionality.

- Mitigation: Abstraction layer isolating exchange-specific code from core business logic. Monitoring systems alerting to API changes or failures. Relationships with exchange developer programs for advance notice of changes. Support for multiple exchanges reducing dependency on any single provider.
- Risk: Security breach compromising client data or exchange credentials.
- Mitigation: HashiCorp Vault integration for secure credential storage. Row-Level Security ensuring complete tenant data isolation. End-to-end encryption for data in transit and at rest. Regular security audits and penetration testing. Compliance preparation for LGPD, SOC 2, and CCSS standards.
- Risk: Scalability limitations as user base grows beyond current architecture capacity.
- Mitigation: Modular architecture designed for microservices migration. Cloud-native infrastructure supporting horizontal scaling. Performance monitoring identifying bottlenecks before they impact users. Documented scalability roadmap with clear triggers for infrastructure upgrades.
- **Legal and Regulatory Risks:**
- Risk: Regulatory changes in cryptocurrency markets requiring significant product adaptations.
- Mitigation: Active monitoring of regulatory developments in Brazil and key markets. Flexible architecture allowing rapid implementation of compliance requirements. Legal counsel specializing in fintech and cryptocurrency regulation. Participation in industry associations influencing regulatory discussions.
- Risk: Licensing or compliance requirements creating barriers to market entry.

- Mitigation: Proactive engagement with regulatory bodies to understand requirements. Compliance-first approach in product development decisions. Budget allocation for legal and compliance expenses. Partnerships with compliance-specialized firms for guidance.
- **Competitive Risks:**
- Risk: Entry of well-funded competitors or established financial institutions into the market.
- Mitigation: First-mover advantage in specialized crypto portfolio management for Brazilian market. Deep domain expertise creating barriers to replication. Strong client relationships and switching costs through white-label integration. Continuous innovation maintaining feature leadership. Focus on underserved market segment (small/medium investment offices) less attractive to large players.
- Risk: Exchange platforms developing native portfolio management features reducing need for third-party solutions.
- Mitigation: Multi-exchange consolidation as core value proposition that single exchanges cannot replicate. Focus on professional/institutional features beyond basic portfolio tracking. White-label customization enabling client branding that exchanges do not offer. Advisory and service layer complementing pure technology offering.
- **Operational Risks:**
- Risk: Key personnel departure impacting development velocity or institutional knowledge.
- Mitigation: Comprehensive technical documentation reducing knowledge concentration. Competitive compensation and equity participation for key team members. Cross-training ensuring multiple team members understand critical systems. Hiring pipeline development for rapid replacement capability.

- Risk: Third-party service failures (Auth0, cloud providers, etc.) disrupting platform availability.
- Mitigation: Selection of enterprise-grade service providers with strong SLAs. Fallback mechanisms for critical services where feasible. Monitoring and alerting for third-party service status. Incident response procedures for rapid user communication during outages.

3 Conclusion

Conclusion

This work presented the development and validation of Vault Wealth, a white-label wealth management platform for cryptocurrency assets designed for investment offices and specialized consultancies. The project addressed a clear market gap: the lack of integrated, professional-grade tools for managing multiple client portfolios distributed across centralized exchanges and on-chain wallets.

Achievement of Objectives:

The general objective of creating and validating a complete computational solution was successfully achieved. A functional MVP was developed encompassing multi-exchange data consolidation, automated buy/sell operation calculations, portfolio rebalancing algorithms, real-time analytical dashboards, and secure credential management through HashiCorp Vault integration.

The specific objectives were met as follows: The platform was implemented using a modern technology stack (React, TypeScript, NestJS, PostgreSQL, Redis, RabbitMQ) with a secure multi-tenant architecture ensuring complete data isolation between organizations. Integration with major cryptocurrency exchanges via API was accomplished, enabling automated portfolio synchronization. The platform was validated with internal users at Vault Capital, processing over 150 real transactions with 99.8% uptime. The revenue model based on monthly subscriptions with dynamic pricing was defined and validated through market conversations. A comprehensive go-to-market strategy encompassing both B2B and B2C channels was developed. Technical and business documentation was prepared for PIPE/FAPESP investment opportunity submission.

Technical Contributions:

The project delivered significant technical contributions including a robust multi-tenant architecture with Row-Level Security for complete data isolation, real-time WebSocket infrastructure with four specialized gateways delivering sub-second latency, intelligent trade execution engine supporting both real and mock trading scenarios, comprehensive testing framework with 634+ lines of test coverage, and 17,622+ lines of production code across 22 NestJS modules.

Business Contributions:

From a business perspective, the project established a validated business model with clear monetization paths through white-label licensing and future B2C subscriptions. The market analysis confirmed significant opportunity in Brazil, where 43% of the population has invested in crypto assets. The competitive analysis revealed a strategic gap that Vault Wealth is uniquely positioned to fill.

Future Projections:

The venture is positioned for growth across multiple dimensions. In the short term (next 6 months), priorities include completing the PIPE Phase 1 funding process, finalizing multi-tenant implementation for external clients, launching controlled pilots with partner investment offices, and expanding exchange integrations to include additional platforms.

In the medium term (6-18 months), the focus shifts to scaling the white-label client base, implementing advanced analytics with machine learning capabilities, developing mobile applications for managers and end clients, achieving SOC 2 Type I certification, and launching the B2C direct offering for individual investors.

In the long term (18+ months), expansion plans include international growth into LATAM markets, advanced compliance modules for multiple jurisdictions, DeFi protocol integration (staking, yield farming, liquidity pools), public API for third-party developers, and potential strategic partnerships or investment opportunities.

Final Considerations:

The development of Vault Wealth demonstrated that it is possible to create a sophisticated, enterprise-grade cryptocurrency portfolio management platform with a lean team and limited resources by leveraging modern technologies, agile methodologies, and continuous user validation. The internal validation at Vault Capital proved invaluable, providing real-world feedback that shaped every aspect of the product.

The cryptocurrency market continues its trajectory of growth and institutionalization, with regulatory frameworks maturing and traditional financial players increasingly entering the space. This macro environment creates favorable conditions for specialized infrastructure solutions like Vault Wealth that bridge the gap between crypto-native technology and traditional financial management practices.

The journey of building Vault Wealth reinforced the importance of resilience and persistence in entrepreneurship. As Churchill noted, when going through difficult times, the only way forward is to keep going. The challenges encountered during development, from technical architecture decisions to market validation pivots, ultimately strengthened both the product and the team.

Vault Wealth stands ready to serve the growing market of investment professionals seeking to offer cryptocurrency services to their clients with the same level of professionalism, security, and efficiency they expect from traditional financial tools.

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